DAKAR BITCOIN HACK BY BTRUST · 2026

Natt ⚡

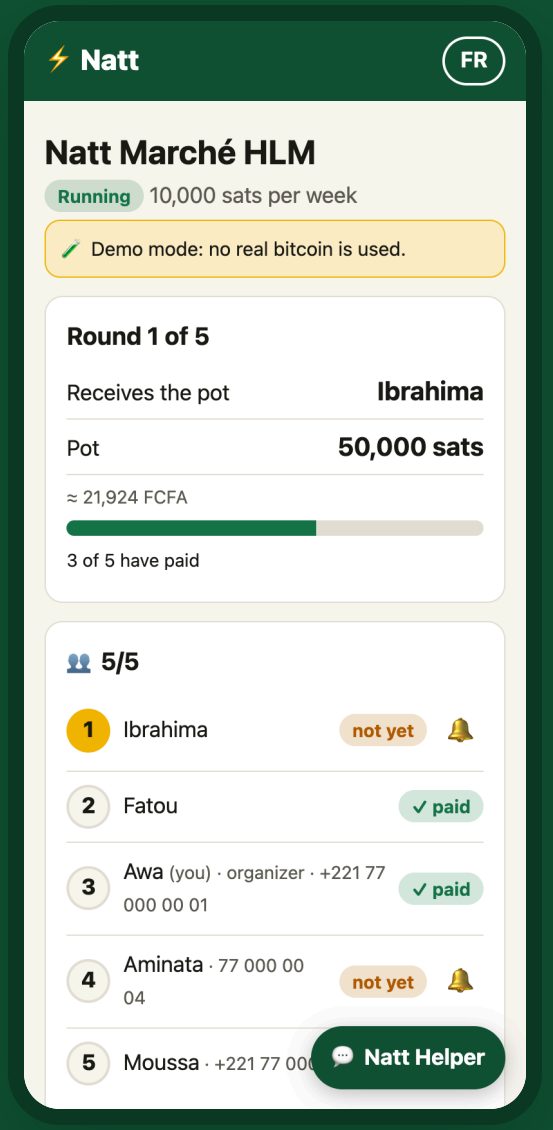
Lightning tontines for West Africa

The rotating savings group people already trust, on Bitcoin Lightning, with tamper-proof Nostr receipts and an AI helper in French, Wolof and English.

Live demo shineyrbl-byte.github.io/natt

Video youtu.be/Wv3Vld1ncZw

Code github.com/shineyrbl-byte/natt



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THE PROBLEM

Savings that run on cash and trust

In a **natt** (Wolof for tontine), every member pays the same amount each week or month, and one member takes the whole pot, until everyone has had a turn.

pot = members × contribution

Market vendors, families and friends use it to save without a bank. Early receivers get an interest-free loan; later ones get forced savings.

💵 Cash carried by hand

The organizer walks around collecting and holding everyone's money.

📒 Records in a notebook

When people disagree about who paid, there's no proof.

🏃 The pot can disappear

If the organizer or a member runs off, the group has no recourse.

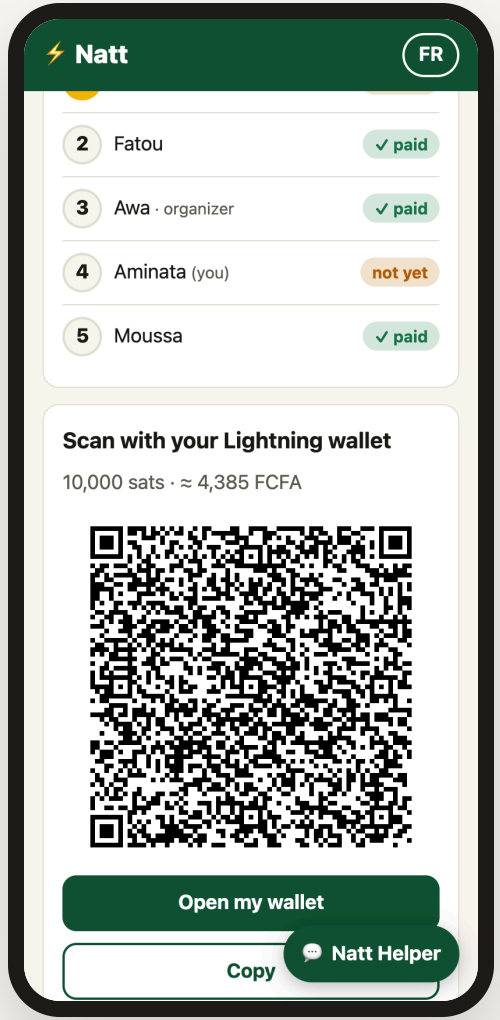
🌍 Family abroad is left out

Sending a small share across borders is slow and expensive.

THE SOLUTION

Keep the natt. Remove the weak points.

1. **Create & invite.** The organizer sets the amount and size; invites go out on WhatsApp.
2. **Fair draw.** The payout order is drawn with a cryptographic random number generator.
3. **Pay by Lightning QR.** From any wallet (Wallet of Satoshi, Phoenix, Blink…), from Dakar or Paris, in seconds.
4. **Guarded payout.** The pot goes only to the member whose turn it is, only after everyone has paid.
5. **Public receipts.** Every payment is a signed Nostr event. Edit or delete one and the chain breaks.



BITCOIN / LIGHTNING WALLET IMPLEMENTATION

A real Lightning wallet behind every group

Member wallet
Scans the QR and pays the BOLT11 invoice

→

Group wallet
Invoice per member per round; payment detected and recorded

→

Payout checks
Right member · all paid · exact amount · not expired · lock

→

Receiver wallet
Paid to their invoice or Lightning address

One wallet interface

`createInvoice` · `isPaid` · `payInvoice` · `getBalance`. The tontine logic never talks to a node directly.

LNbits: real sats

The invoice key creates invoices; the admin key, which can spend, is used only for payouts.

Demo wallet

Signs real, decodable BOLT11 invoices (secp256k1), so judges can try it with no node and no funds.

Lightning addresses

LNURL-pay payouts to `name@wallet`, with private-network blocking and an amount check on the returned invoice.

No double payout

A `collecting → paying → paid` state lock inside a database transaction.

Sats first

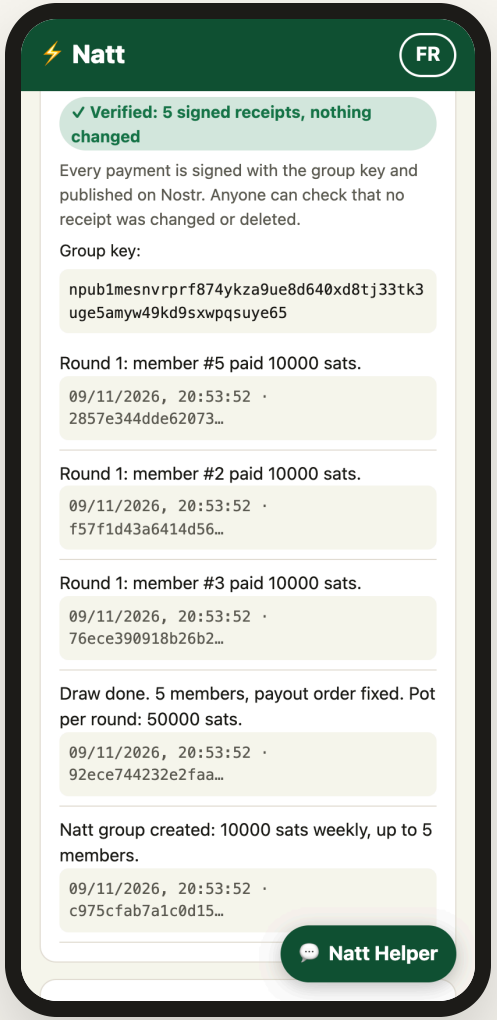
Amounts are stored in sats. FCFA is display only, from a live BTC/EUR price and the fixed euro peg.

NOBODY CAN CHEAT

A public ledger anyone can check

- Each group has its **own Nostr key**, derived from a server secret.
- Group creation, the draw, every payment and every payout become **signed kind 1 events**.
- Each event **links to the previous one** with an `e` tag, so the history forms a chain.
- The app re-checks every signature and link, and shows ✓ **Verified**, or exactly what broke.
- Anyone with the group code can audit it. Publishing to public relays is optional.

"Trust the organizer" becomes "check the receipts".



MADE FOR THE PEOPLE WHO RUN NATTS

Local language, local tools

💬 Natt Helper (AI)

"When is my turn?", "Who hasn't paid?", answered in simple words with sats and FCFA. Uses Claude when a key is set, built-in answers offline.

🗣️ French, Wolof, English

One tap switches language, for the app and the helper. French by default.

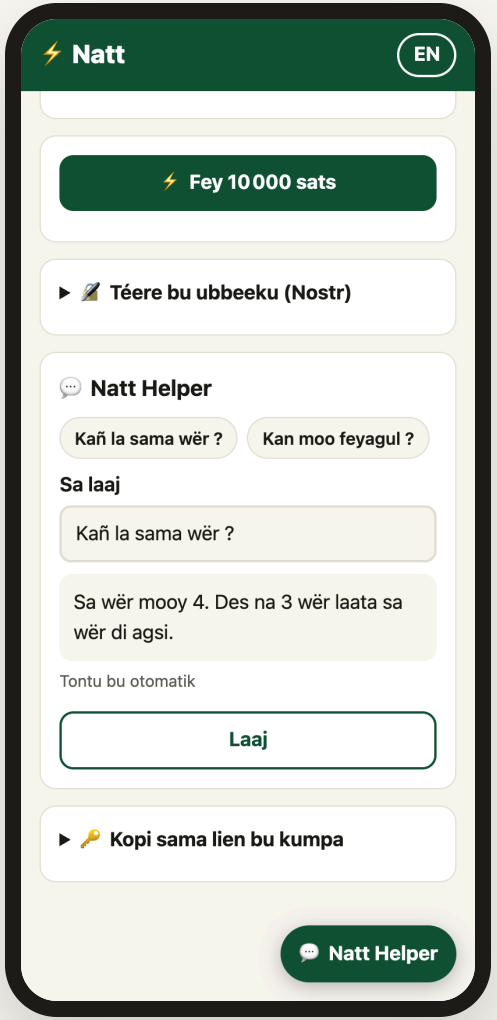
🔔 WhatsApp reminders

The organizer taps a bell and WhatsApp opens with a polite reminder. Natt never sends on its own.

📱 Built for real phones

Light installable web app, no framework, large buttons, works on slow networks.

The AI only sees first names and paid / unpaid status: no phone numbers, no tools, and it cannot move money.



SECURITY BY DESIGN

Written threat model, 11 threats, each mitigated

Money

- Payout guards and state lock
- Spending key used only for payouts
- Exact-amount and expiry checks

Identity

- Secret member tokens, stored hashed
- Constant-time comparison
- Sent in headers, not URLs

Privacy

- Phones visible only to the organizer
- Logs redact phones, invoices, keys
- AI gets a summary, no phones

Web app

- Strict Content Security Policy
- No `innerHTML` with user data
- Rate limiting, strict input validation

Network

- Lightning address fetches block private hosts
- No redirects followed
- Timeouts on Lightning, price and AI calls

Honest trade-off

- The group wallet holds the pot between collection and payout
- Next: hold invoices to shrink that

WORKING MVP

Try it now, or run it with one command

Live	1 command	39	3
shineyrbl-byte.github.io/natt	docker compose up · npm run demo	automated tests, CI on every push	languages: FR · WO · EN

On the live demo

- Tap a member under **Try the demo** to join a seeded market group of five vendors.
- Pay with the demo wallet, claim the pot as the member whose turn it is, check the ledger.
- No real bitcoin is used. Switch to real sats with three LNbits settings.

Built with

Node.js · Express · SQLite · Lightning · BOLT11 · LNURL · LNbits · Nostr · nostr-tools · noble-curves · Claude · PWA · Docker · GitHub Actions · WebAssembly · GitHub Pages

WHAT'S NEXT

From demo to Marché HLM

- **Less custody:** hold invoices, so a payment only settles when the payout happens.
- **Nostr login** (NIP-07 / Nostr Wallet Connect) and encrypted group chat.
- **Scheduled reminders**, including SMS for members without WhatsApp.
- **Native-speaker review** of the Wolof text.
- **Pilot** with organizers at Marché HLM, Dakar. One organizer runs 10 to 20 members, so ten organizers means about 200 people paying by Lightning every week.

The tontine you know, without the cash and without the doubt. Jërëjëf.